WHY DO I NEED TO PIERCE THE FILM WHEN HEATING READY MEALS?

As microwaves heat water molecules in food, they expand to become steam. Piercing the film lets steam escape the container, which might otherwise explode.

How a microwave works

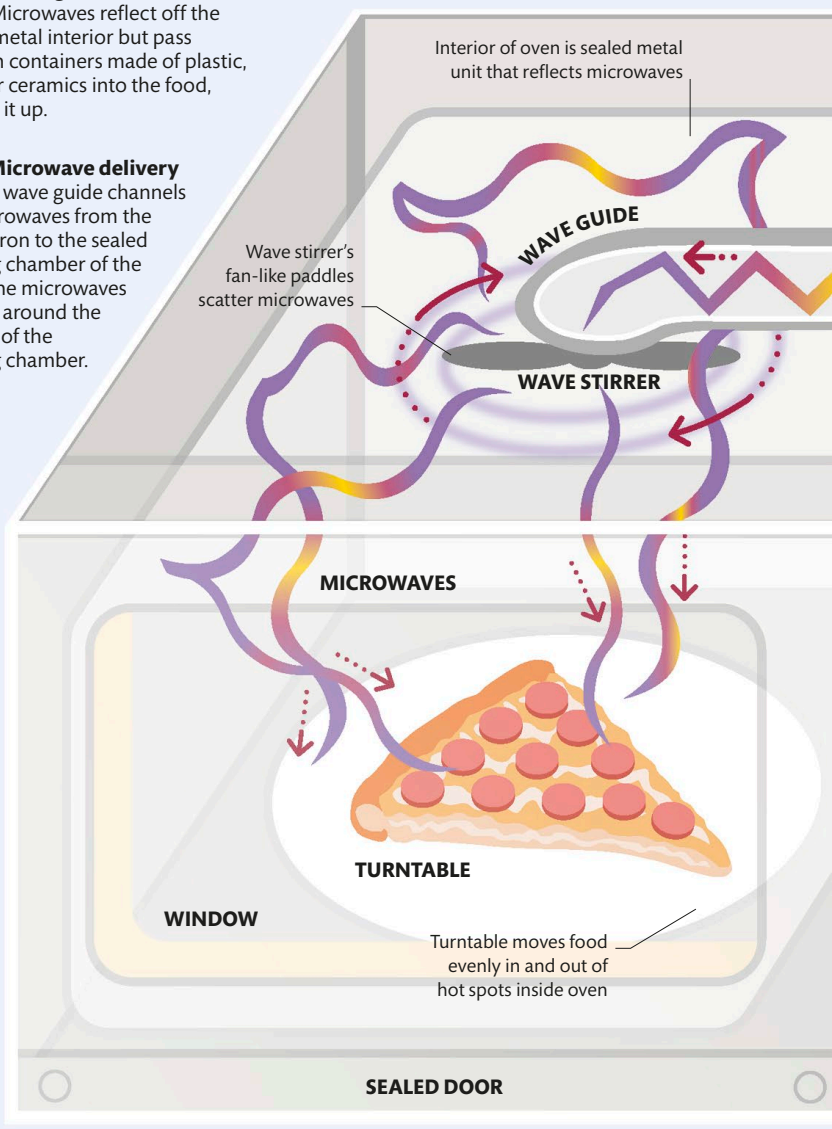
A home microwave oven uses AC power to power a magnetron. This device uses interacting electric and magnetic fields to produce microwaves, which oscillate and reverse their electric field several billion times a second. The microwaves are directed into the oven's cooking chamber—a sealed metal box—where they bounce around, striking and exciting molecules in food, which heat up as a result.

2 Microwave creation
The magnetron generates microwaves oscillating at a frequency of 2.45GHz (2.45 billion times per second).

1 Control settings
The user selects power and time settings, often using a touchscreen panel. Safety switches in the door cut power if the door is opened while the oven is working.

4 Heating food
Microwaves reflect off the oven's metal interior but pass through containers made of plastic, glass, or ceramics into the food, heating it up.

3 Microwave delivery
A wave guide channels the microwaves from the magnetron to the sealed cooking chamber of the oven. The microwaves bounce around the interior of the cooking chamber.



Microwave ovens

Microwaves are a type of energy found on the electromagnetic spectrum between infrared and radio waves (see pp.136–137). They pass through many, but not all, materials and can penetrate food to agitate water and fat molecules, generating heat so that foods are cooked evenly and more quickly than in a conventional oven.



THE **FIRST COMMERCIAL MICROWAVE** OVEN WAS **5½ FT (1.7 M) TALL**